Homework 2
26 January 2025
Total Pages: 10

Justus Languell
135007144
ENGR 216 466

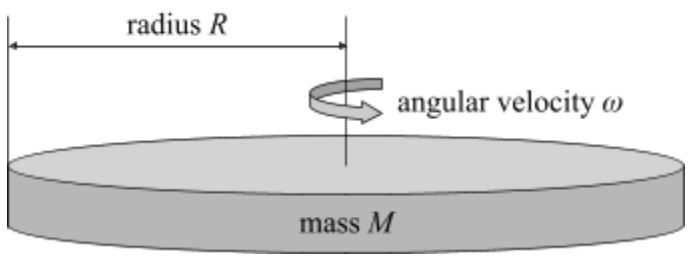
Question 1

Given:
The mass M, radius R, and angular velocity ω, of a rotating disk, where

- $M = 2.50 \pm 0.020\ kg$
- $R = 0.180 \pm 0.0030\ m$
- $\omega = 17.5 \pm 0.250\ rad/s$

Find:
The uncertainty of the angular momentum L.

Diagram:



Theory:
Angular momentum $L = \frac{1}{2}MR^2\omega$

Uncertainty of function q with three parameters x, y, and z

$$\delta q = \sqrt{\left(\frac{\partial q}{\partial x}\delta x\right)^2 + \left(\frac{\partial q}{\partial y}\delta y\right)^2 + \left(\frac{\partial q}{\partial z}\delta z\right)^2}$$

Assumptions:
Since the number of samples is very large, it is safe to assume that this represents a population.

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Solution:

The angular velocity can be calculated to be as follows

$$L = \frac{1}{2}MR^2\omega = (2.5)(0.180)^2(17.5) = \boxed{0.708\ \frac{kg \cdot m^2}{s}}$$

The uncertainty for angular velocity is expressed as

$$\delta L = \sqrt{\left(\frac{\partial L}{\partial M}\delta M\right)^2 + \left(\frac{\partial L}{\partial R}\delta R\right)^2 + \left(\frac{\partial L}{\partial \omega}\delta\omega\right)^2}$$

The partial derivatives required to solve for δL are

$$\frac{\partial L}{\partial M} = \frac{1}{2}R^2\omega = \frac{(0.180)^2(17.5)}{2}$$
$$\frac{\partial L}{\partial R} = MR\omega = (2.50)(0.180)(17.5)$$
$$\frac{\partial L}{\partial \omega} = \frac{1}{2}MR^2 = \frac{(2.50)(0.180)^2}{2}$$

And we can plug this in to our main expression for δL below

$$\delta L = \sqrt{\left(\frac{(0.180)^2(17.5)}{2} \cdot \delta M\right)^2 + \left((2.50)(0.180)(17.5) \cdot \delta R\right)^2 + \left(\frac{(2.50)(0.180)^2}{2} \cdot \delta\omega\right)^2}$$

Plugging in individual uncertainties we get

$$\delta L = \sqrt{\left(\frac{(0.180)^2(17.5)}{2}(0.020)\right)^2 + \left((2.50)(0.180)(17.5)(0.0030)\right)^2 + \left(\frac{(2.50)(0.180)^2}{2}(0.250)\right)^2}$$
$$= \boxed{0.0263\frac{kg \cdot m^2}{s}}$$

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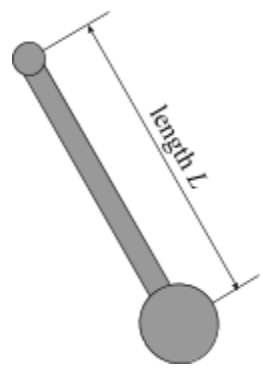
Question 2

Given:
The length of a pendulum $L = 0.75 \pm 0.011$ m.

Find:
a. The predicted value of T?
b. Is a measured value of $T = 1.75 \pm 0.010$ s consistent with the prediction of part (a)?

Diagram:



Theory:

Period of a pendulum $T = 2\pi\sqrt{\frac{L}{g}}$

Uncertainty of function f of a single parameter x $\delta f = \frac{df}{dx}\delta x$

Assumptions:
The gravitational acceleration constant g is exactly $g = 9.81$ m/s^2.

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Solution:

a. The value for the expected period can be calculated to be as follows

$$T = 2\pi\sqrt{\frac{L}{g}} = 2\pi\sqrt{\frac{0.75}{9.81}} = \boxed{1.74\ s}$$

The uncertainty of T can also be expressed as

$$\delta T = \frac{dT}{dL}\delta L = \frac{\pi}{g\sqrt{\frac{L}{g}}}\delta L = \frac{\pi}{9.81\sqrt{\frac{0.75}{9.81}}}(0.011) = \boxed{0.013\ s}$$

b. The measured value of $T = 1.75 \pm 0.011\ s$ is consistent with the theoretical value because it satisfies the following condition

$$1.75 - 0.011 \leq 1.737 \leq 1.75 + 0.011$$

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Question 3

Given:

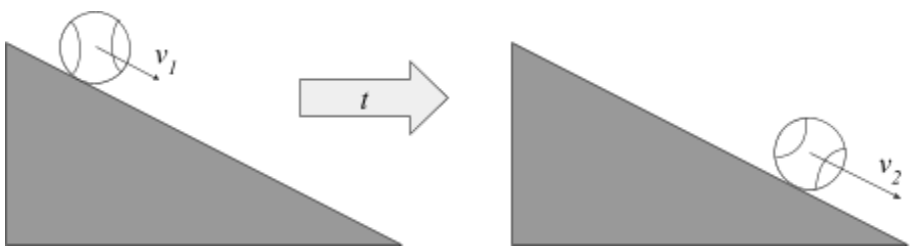
For a baseball rolling down a ramp, you have velocity measurements at two points (v_1 and v_2) and the time t it between them, with measurements being

$v_1 = 3.54 \pm 0.10\ m/s$
$v_2 = 8.16 \pm 0.10\ m/s$
$t = 2.79 \pm 0.10\ s$

Find:

a. The acceleration, a.
b. Its uncertainty δa.

Diagram:



Theory:

Acceleration $$a = \frac{v_2 - v_1}{t}$$

Uncertainty of function q with three parameters x, y, and z

$$\delta q = \sqrt{\left(\frac{\partial q}{\partial x}\delta x\right)^2 + \left(\frac{\partial q}{\partial y}\delta y\right)^2 + \left(\frac{\partial q}{\partial z}\delta z\right)^2}$$

Assumptions:

The ball rolls down the ramp ideally and does not slip.

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Solution:

The acceleration can be calculated to be as follows

$$a = \frac{v_2 - v_1}{t} = \frac{8.16 - 3.54}{2.79} = \boxed{1.66\ \frac{m}{s^2}}$$

The uncertainty for acceleration is expressed as

$$\delta a = \sqrt{\left(\frac{\partial a}{\partial v_1}\delta v_1\right)^2 + \left(\frac{\partial a}{\partial v_2}\delta v_2\right)^2 + \left(\frac{\partial a}{\partial t}\delta t\right)^2}$$

The partial derivatives required to solve for δL are

$$\frac{\partial a}{\partial v_1} = -\frac{1}{t} = -0.3584$$

$$\frac{\partial a}{\partial v_2} = \frac{1}{t} = 0.3584$$

$$\frac{\partial a}{\partial t} = -\frac{v_2 - v_1}{t^2} = -\frac{8.16 - 3.54}{(2.79)^2} = -0.5935$$

And we can plug this in to our main expression for δa below

$$\delta a = \sqrt{((-0.3584)\delta v_1)^2 + ((0.3584)\delta v_2)^2 + ((-0.5935)\delta t)^2}$$

Plugging in individual uncertainties we get

$$\delta a = \sqrt{((-0.3584)(0.1))^2 + ((0.3584)(0.1))^2 + ((-0.5935)(0.1))^2} = \boxed{0.078\ \frac{m}{s^2}}$$

The measured value of $a = 1.85 \pm 0.10\ m/s^2$ is not consistent with the theoretical value because it does not satisfy the following condition

$$1.85 - 0.10 \leq 1.656 \leq 1.85 + 0.10 \equiv \perp$$

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Question 4

Given:
The thickness of a textbook is 1.25 in, with uncertainty of ± 0.0050 in.
Let *N* be the number of pages, 437.

Find:

a. The thickness of one page.
b. The number of (whole) textbooks that you need to measure together if you want to know the thickness of 1 page with an uncertainty of only $5.0 * 10^{-6}$ in.

Diagram:

Theory:

Uncertainty of function *f* of a single parameter *x* $\delta f = \frac{df}{dx}\delta x$

Assumptions:

- We do not need to account for the cover or any other dimensions.
- The pages are perfectly flat.

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Solution:

a. Width of a single page $w = \frac{T}{N} = \frac{1.25}{437} = \boxed{0.0028\ in}$

b. The uncertainty of the width of a page given the uncertainty of the thickness and number of pages

$$\delta w = \frac{1}{n}\delta T$$

We can solve this for the individual uncertainty for the width of each page

$$\delta w = \frac{1}{437}(0.0050) = \boxed{1.14 \cdot 10^{-5}}$$

Rearrange in terms of number of books, *B*

$$n\delta w = \delta T$$

$$BN = \frac{\delta T}{\delta w}$$

$$B = \frac{\delta T}{N \cdot \delta w}$$

And solve for minimum number of books to achieve a desired uncertainty

$$B = \frac{0.0050}{437 \cdot 5 \cdot 10^{-6}} = 2.28$$

And perform ceil to convert to a whole number

$$ceil(2.28) = \boxed{3\ books}$$

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Question 5

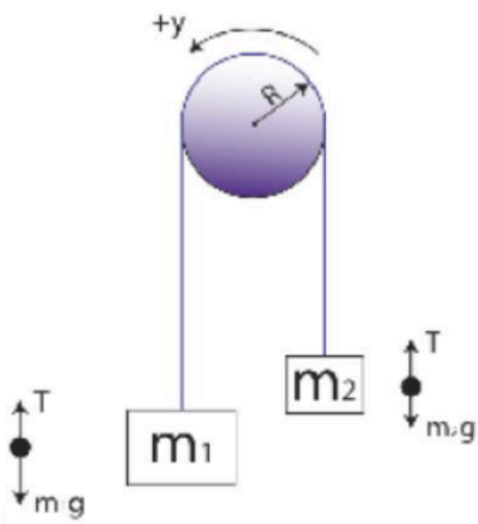
Given:

- Masses are $m_1 = 102 \pm 1.0\ g$ and $m_2 = 86 \pm 0.90\ g$.

Find:

a. an equation for the uncertainty in the expected acceleration in terms of masses and their uncertainties.
b. calculate $a \pm \delta a$ for the given numbers

Diagram:



Theory:

Acceleration $$a = g\frac{m_1 - m_2}{m_1 + m_2}$$

Uncertainty of function q with three parameters x, y, and z

$$\delta q = \sqrt{\left(\frac{\partial q}{\partial x}\delta x\right)^2 + \left(\frac{\partial q}{\partial y}\delta y\right)^2 + \left(\frac{\partial q}{\partial z}\delta z\right)^2}$$

Assumptions:

The gravitational acceleration constant g is exactly $g = 9.81\ m/s^2$.

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Solution:

a. The uncertainty for acceleration is expressed as

$$\delta a = \sqrt{\left(\frac{\partial a}{\partial m_1}\delta m_1\right)^2 + \left(\frac{\partial a}{\partial m_2}\delta m_2\right)^2}$$

The partial derivatives required to solve for δa are

$$\frac{\partial a}{\partial m_1} = \frac{2gm_2}{(m_1+m_2)^2}$$
$$\frac{\partial a}{\partial m_2} = -\frac{2gm_1}{(m_2+m_1)^2}$$

And we can plug this in to our main expression for δa below to get an equation for the uncertainty in the expected acceleration in terms of masses and their uncertainties

$$\boxed{\delta a = \sqrt{\left(\frac{2gm_2}{(m_1+m_2)^2}\delta m_1\right)^2 + \left(-\frac{2gm_1}{(m_2+m_1)^2}\delta m_2\right)^2}}$$

b. Plugging in individual uncertainties we get

$$\delta a = \sqrt{\left(\frac{2(9.81)(86)}{(102+86)^2}(1.0)\right)^2 + \left(-\frac{2(9.81)(102)}{(86+102)}(0.9)\right)^2} = 0.07\,\frac{m}{s^2}$$

And calculating for a we get

$$a = g\frac{m_1-m_2}{m_1+m_2} = (9.81)\frac{102-86}{86+102} = 0.8349$$

Therefore

$$\boxed{a \pm \delta a = 0.8346 \pm 0.07}$$